

Uplift

AI-Powered Student Success Assistant

Early warning, similar past students and supportive action plans for Open University module teams: hybrid RAG, a multi-agent workflow, calibrated prediction models and LLM-as-judge evaluation.

Capstone project · Education domain · Open University Learning Analytics Dataset (OULAD)

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PROBLEM STATEMENT

Advisors cannot read thousands of records in time

Students who disengage or fall behind are found too late, and dashboards say who, not why or what to do.

- Module teams hold rich data: profiles, assessment results, daily online activity, registration history, outcomes.
- Reviewing it by hand is slow and inconsistent; warning signs are spread across tables and weeks.
- By the time a student formally withdraws, it is too late to help: **75%** of withdrawals in the test presentation happened before week 15.
- Teams need early, explainable, prioritised and supportive guidance, with evidence.

Q1 **Has a student with similar patterns struggled or withdrawn before?**

Hybrid RAG over historical cases

Q2 **What factors are associated with poor performance or withdrawal?**

Explainable models and insights

Q3 **Which students should be prioritised for support?**

Capacity-aware ranking and agent plans

BUSINESS NEED

Reach the right students early, with reasons

Capacity is limited, so the top of the list must be right, and every recommendation must be defensible.

25,204

students still enrolled at week 15 (all modules)

11.3%

of them will still withdraw: the students an early warning can help

98%

of students contacted from the top 10% later fail or withdraw (random: 34%)

50%

of withdrawals are in the High tier, which flags 24% of students

- **Early**: warnings from week 5, not after the student has gone
- **Explainable**: drivers per student and what happened to similar students
- **Capacity-aware**: a ranked list for the time advisors actually have
- **Supportive and safe**: never punitive, no sensitive attributes, human review
- **Measurable**: every model and AI output evaluated, cost tracked

SOLUTION OVERVIEW

One console, six capabilities, as of a course week

Built at weeks 5, 9, 15, 22; only students still enrolled are scored. The LLM narrates, code decides facts.

01

Early warning

Calibrated withdrawal risk, tier, TreeSHAP drivers and outcome probabilities

02

Who to contact first

Ranking learned for advisor capacity, with the reasons for each student

03

Similar students (RAG)

Hybrid search over past cohorts by student or plain English, with metadata filters

04

Support plan (agents)

LangGraph agents: performance, risk, intervention, critic; outreach draft

05

Copilot

Questions answered by tool calls over the same computed functions, streamed and checked

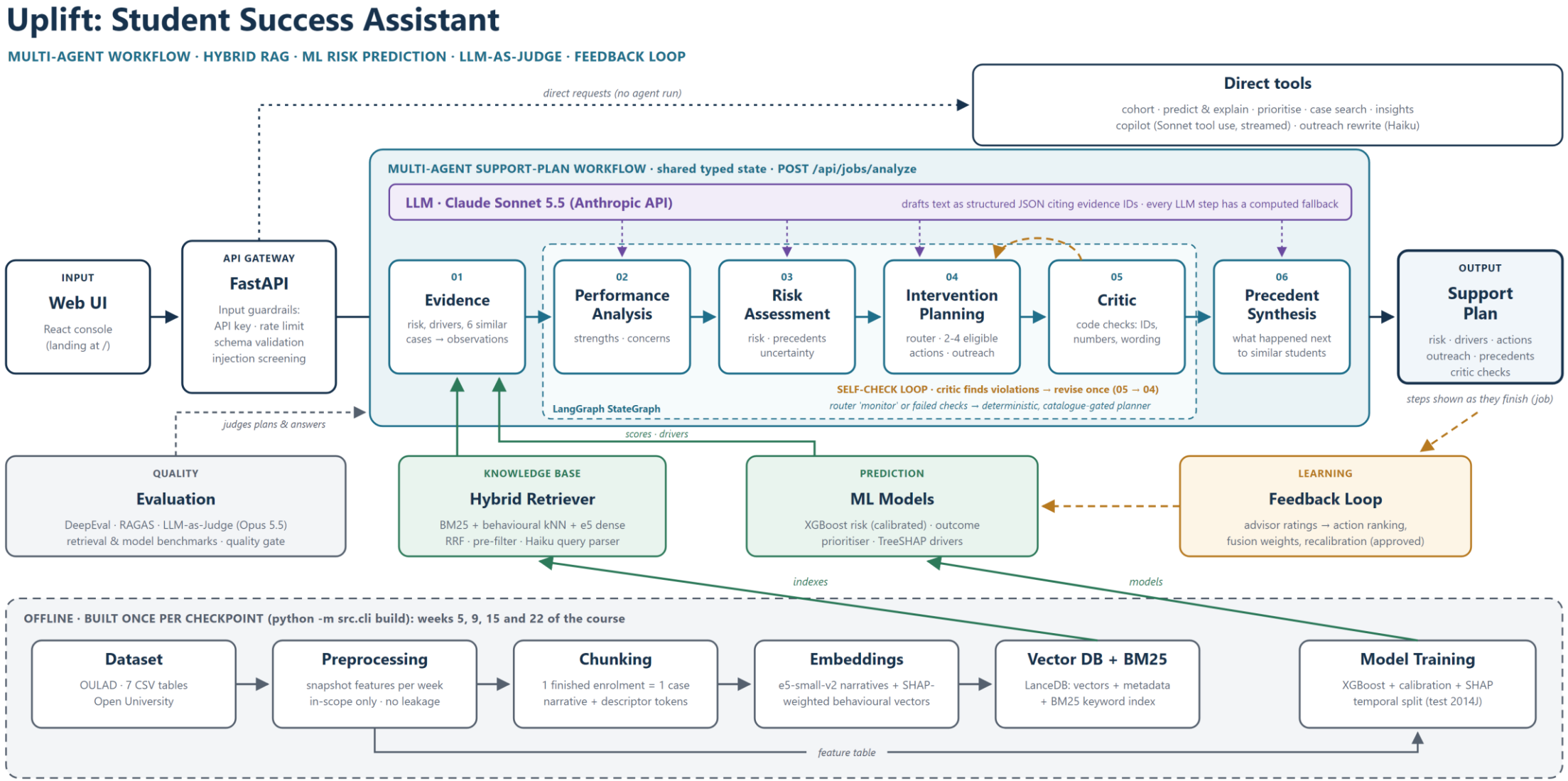
06

Insights and evaluation

Factors, trends, fairness; DeepEval, RAGAS, judge; approved updates

ARCHITECTURE

Offline build per checkpoint, online agents behind FastAPI

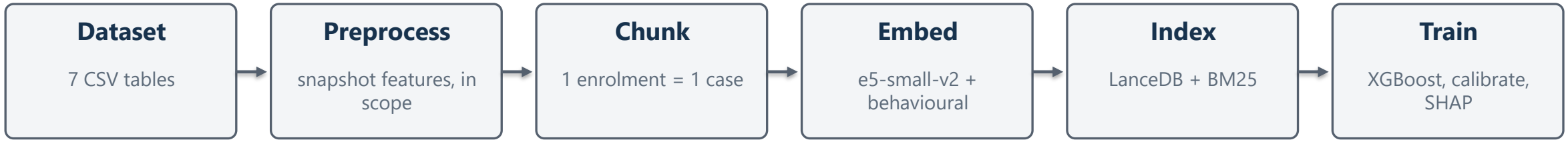


DATA FLOW

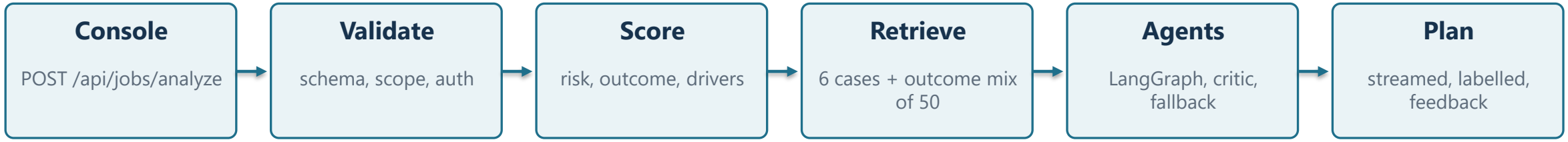
From seven OULAD tables to a checked support plan

Only data dated before the checkpoint is used, and only students still enrolled are scored.

OFFLINE · python -m src.cli build



ONLINE · draft a support plan for one student



The leakage rule

Scoring every enrolment, including students who had already left, gives ROC-AUC **0.749**. Scoring only students still enrolled gives **0.726**: lower, but it is the number an advisor actually gets. `date_unregistration` defines scope only; it is never a feature.

TECHNOLOGY STACK

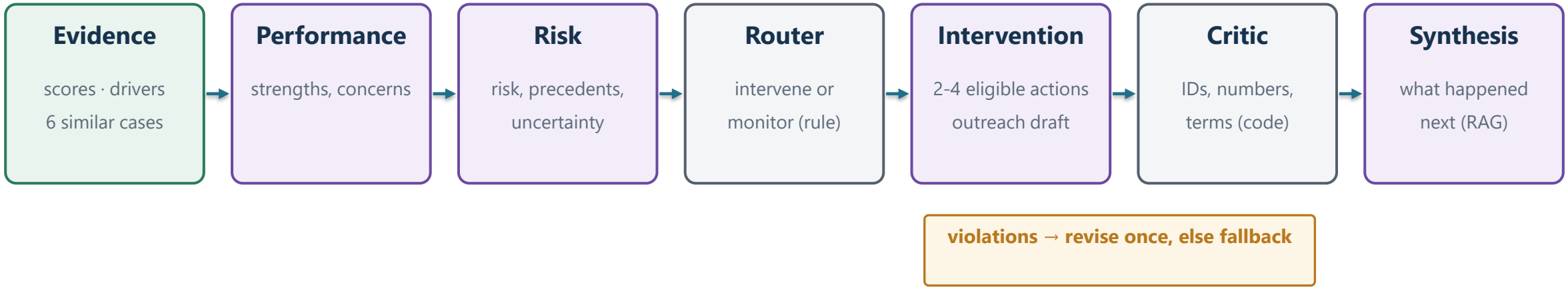
Each choice was measured or justified against an alternative

Layer	Choice	Why
LLM	Claude: claude-haiku-5-5 · claude-sonnet-5-5 · claude-opus-5-5 (judge)	Measured per task: small model parses as well; main model writes most faithfully; different judge
Embeddings	intfloat/e5-small-v2	Best of five models on precedent quality at MiniLM latency
Vector DB	LanceDB (embedded)	No server; metadata pre-filter; exact kNN; parity with brute force
Agents	LangGraph state graph	Explicit router, critic and revision; parity with plain Python
Prediction	XGBoost + calibration + TreeSHAP	ROC-AUC 0.726 vs 0.703 logistic; exact explanations
Retrieval	BM25 + behavioural kNN + dense, RRF	Numbers need numeric similarity; text needs keywords and meaning
API	FastAPI + Uvicorn, pydantic	Typed validation, OpenAPI docs, streaming, background jobs
UI	React 19 + TypeScript (Vite)	Beyond Streamlit: URLs, virtual tables, streaming; served by FastAPI
Evaluation	DeepEval · RAGAS · LLM judge · pytest	Each layer catches a different failure

AGENT WORKFLOW

Agents on a shared state; code checks every draft

Agents cite numbered evidence; the critic rejects unknown IDs, ineligible actions, ungrounded numbers and unsafe words.



Purple = Claude drafts (structured JSON output)

Grey = code only, deterministic

Green = ML models and hybrid retriever

Every LLM step has a computed fallback: the workflow runs with the LLM switched off.

100%

plans pass every guardrail

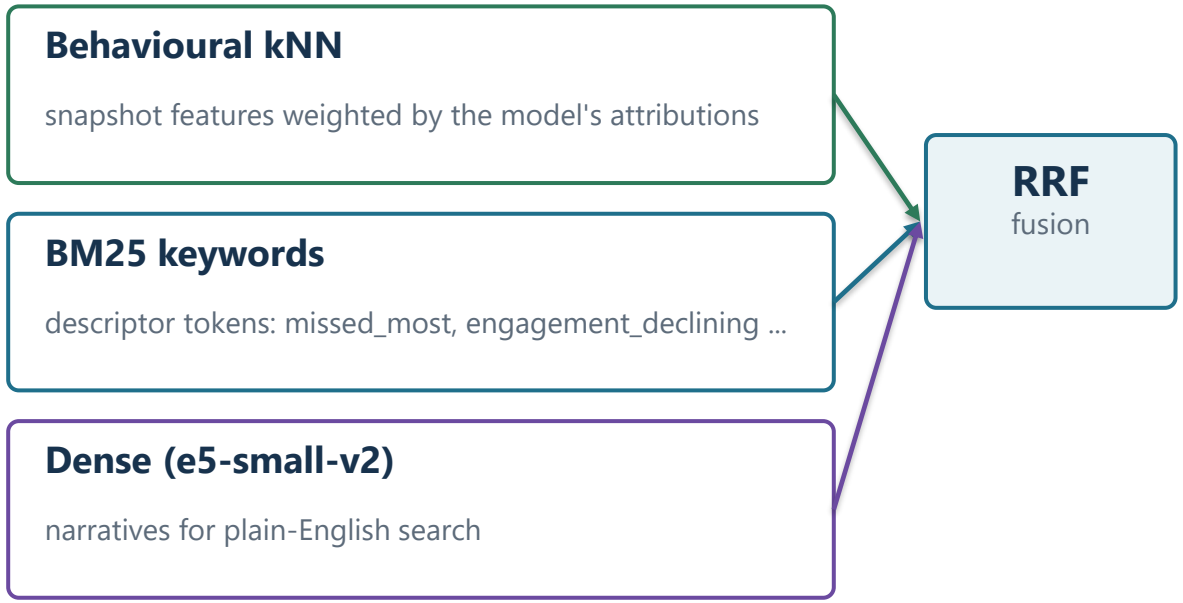
96%

judge pass rate (claude-opus-5-5)

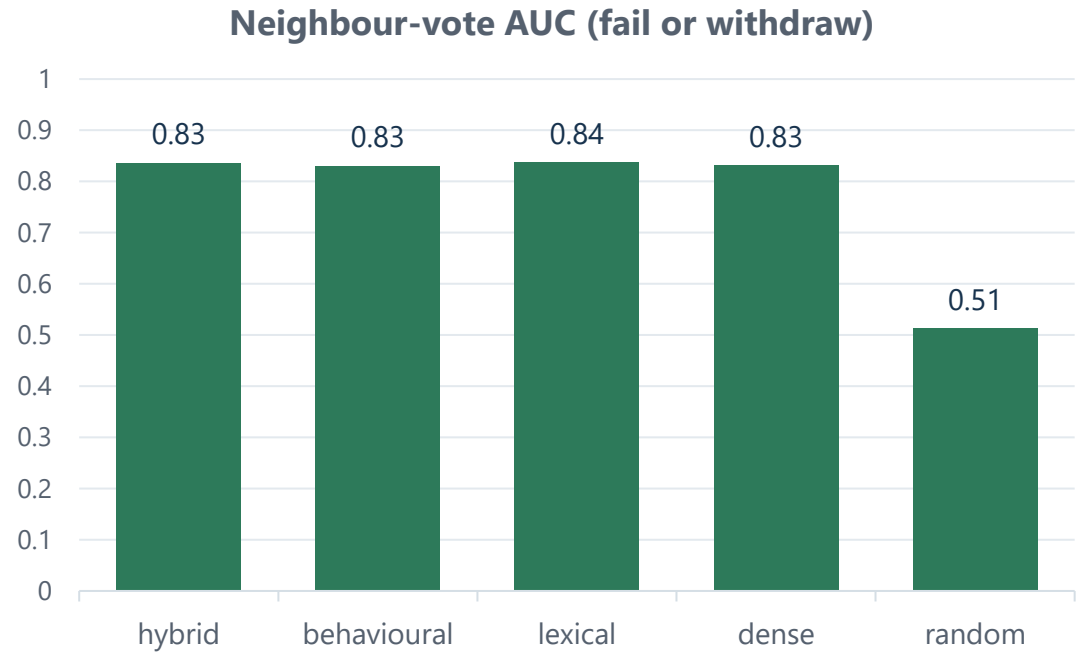
HYBRID RAG

Three channels, fused with RRF, filtered in LanceDB

Earlier presentations only, never the student's own rows; one document per enrolment; numbers banded, not embedded.



- Metadata filters: course, region, education, age band, credits, previous attempts, outcome
- Plain English understood as filters + behaviours (Claude Haiku, rule fallback)
- Context: 6 closest stories; outcome mix of the 50 nearest



Do the retrieved students' real outcomes predict this student's? 600 test students vs earlier cohorts (not circular).

0.862

precision@10, plain-English case search (AI parser)

PREDICTION MODELS

Calibrated risk that holds on a later, untouched cohort

Trained on 2013B + 2013J, tuned on 2014B, tested once on 2014J.

0.726

ROC-AUC at week 15 (95% CI 0.708 to 0.740)

0.009

calibration error: a 20% risk means about 20%

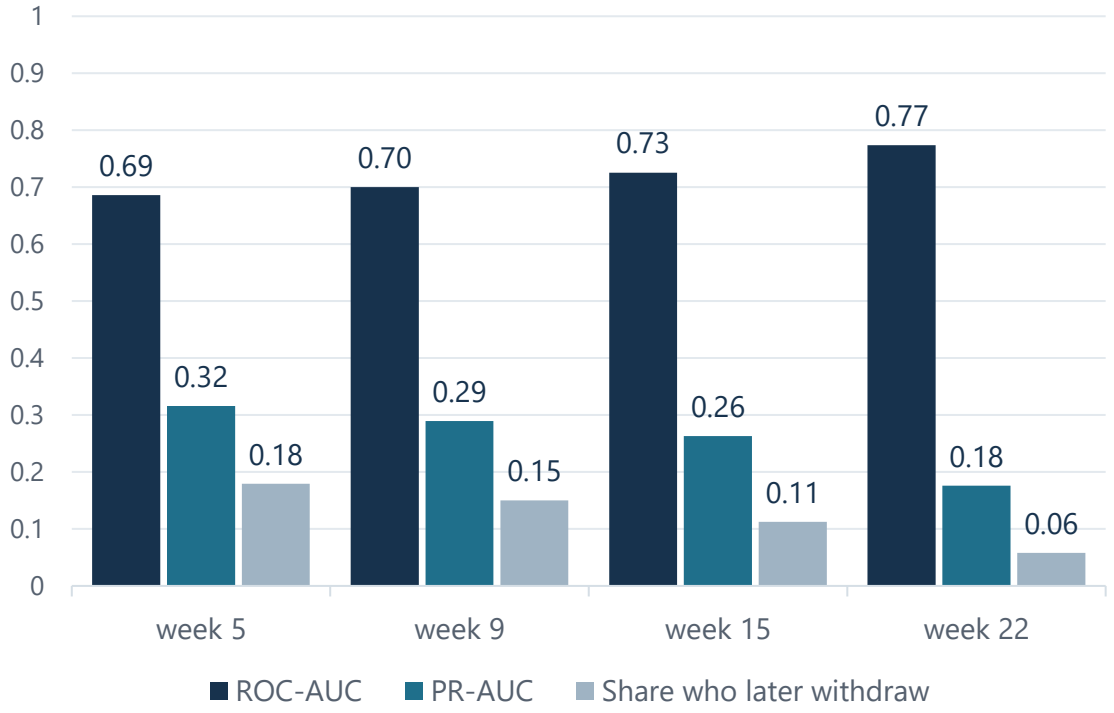
0.703

logistic regression baseline

0.263

PR-AUC vs base rate 0.113

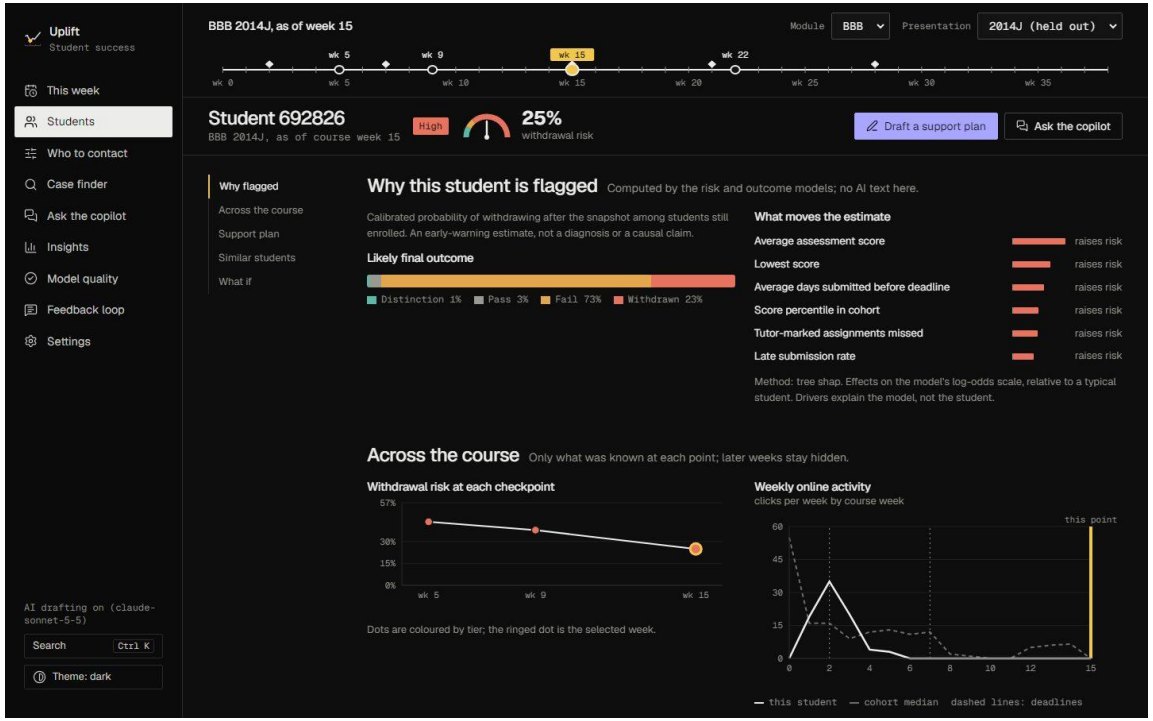
Tier	Share of students	Observed withdrawal rate	Share of withdrawals
High	24%	23.6%	50%
Medium	28%	11.7%	29%
Low	48%	4.9%	21%



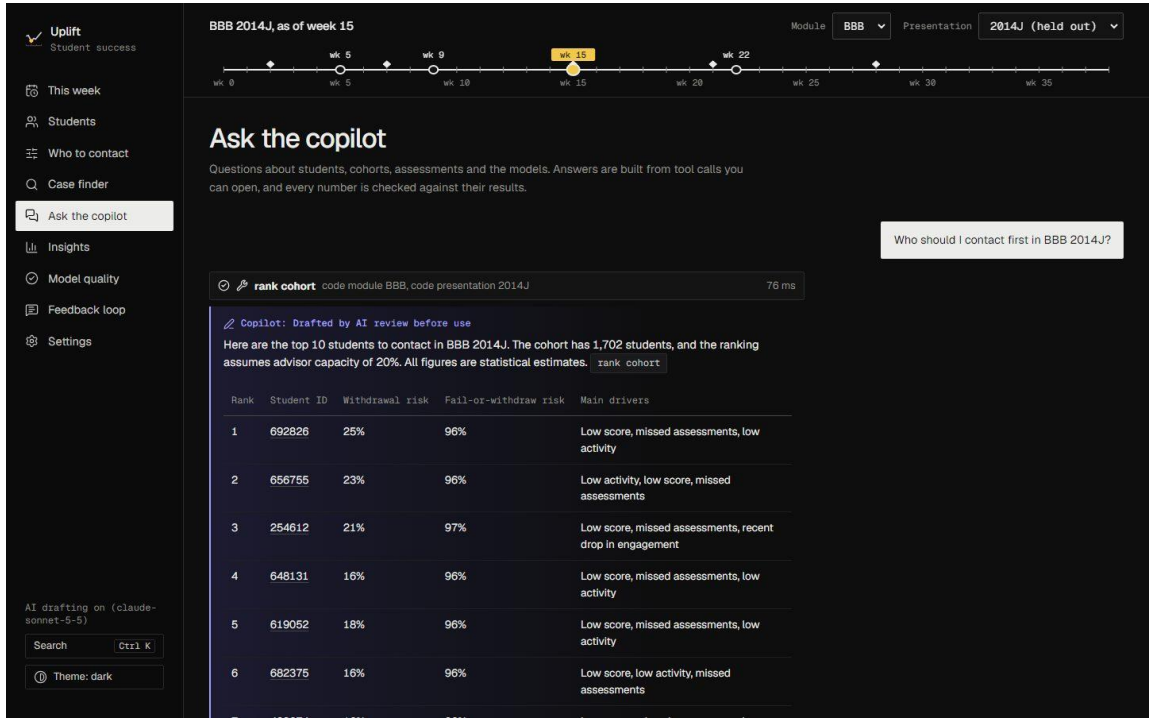
Outcome model macro-F1 0.485 (logistic 0.460); P(fail or withdraw) ROC-AUC 0.852. Ranking precision at 10%: 98%.

DEMO

Live: week, student, support plan, case search, copilot



Student case file: why flagged, risk across the course, support plan



Copilot: tool calls, streamed answer, every number checked

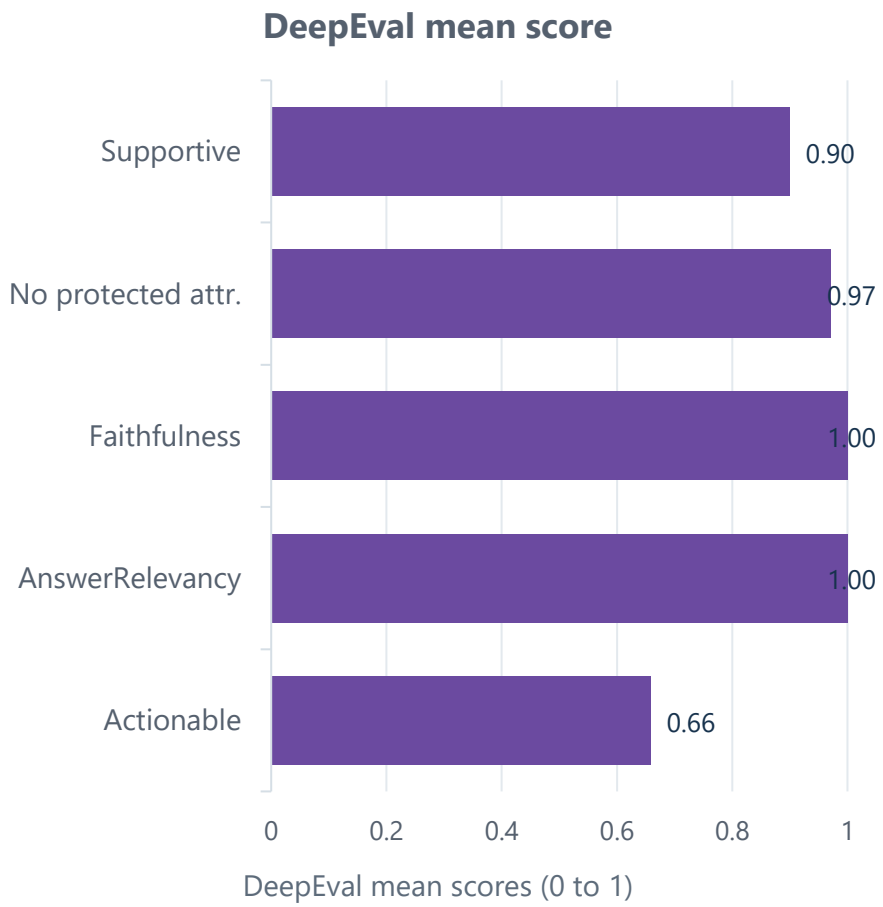
1. uvicorn src.main:app --reload → http://127.0.0.1:8000
2. This week: course clock, week 5 → 15, who to reach first
3. Student: drivers, risk trail, Draft a support plan (agent steps), rewrite as SMS
4. Case finder: plain-English search
5. Copilot: 'Who should I contact first in BBB 2014J?'

EVALUATION RESULTS

Five layers of evaluation, each catching what the others miss

AI evaluation: generators claude-sonnet-5-5 / claude-haiku-5-5; judge claude-opus-5-5; release gate passed.

Layer	Measure	Result
1 Code checks	plans passing every guardrail	100%
2 LLM-as-judge	judge pass rate · AI plan preferred (order-swapped A/B)	96% · 89%
3 DeepEval	faithfulness · answer relevancy · actionable (pass rate)	100% · 100% · 75%
4 RAGAS	faithfulness of precedent synthesis (hallucination rate)	0.964 (3.6%)
5 Copilot	golden questions correct · adversarial prompts handled safely	100% · 88%
Retrieval	neighbour-vote AUC hybrid vs random	0.835 vs 0.512
Tests	pytest suite: no network, no LLM	139 tests



RESPONSIBLE AI

Supportive by design, checked in code, fair by audit

Input guardrails

API key and admin key
Rate limit, request-size limit
Strict schemas and vocabularies
Prompt-injection screening

Output guardrails

IDs must exist, numbers must be in the evidence
Only eligible, supportive actions
No protected, punitive, clinical or stigma terms
Revise once, else computed fallback

Human in the loop

AI text labelled; advisor edits every message
Never for grading, admissions or sanctions
Feedback changes nothing until approved

- **Sensitive attributes** (gender, region, deprivation, age, education, disability) are never model inputs and never sent to the LLM; adding them would change ROC-AUC by only 0.002.
- **Fairness audit**: each group gap is tested against the gap random labels would give at the same group sizes (permutation test, overall and within module).
- Region and deprivation gaps are within chance. The disability gap (0.23) is real and reported, with a per-group threshold review for the equity office, not applied automatically.

CHALLENGES AND LEARNINGS

What was hard, and what changed because of it

Challenge	What we did	Learning
Inflated accuracy (0.90 AUC) from scoring students who had left	Score in-scope students only; temporal split	Evaluate the decision the user actually faces
Numbers in text made keyword and dense search random	Band numbers into descriptors; numeric similarity in its own channel	Hybrid retrieval must match the data type
LLM drafts invented numbers and IDs	Evidence IDs, code checks, one revision, fallback	Let the LLM narrate and select; code decides facts
Self-grading and position bias	Different judge model, computed verdict, order-swapped A/B, human review	Evaluate the evaluator
Validation gains that did not hold on test	Rejected the feature set instead of tuning on test	Keep the test set untouched
RAGAS dependencies broke the pinned service	RAGAS in its own environment	Isolate evaluation tooling



Thank you

Questions, or a live look at any part of Uplift.

Run it: uvicorn src.main:app --reload → http://127.0.0.1:8000

Architecture_Diagram.pdf · Design_Document.pdf · API_Documentation.pdf · Evaluation_Report.pdf · Dataset_Details.pdf

API reference live at /api/docs · model card: docs/MODEL_CARD.md